

# Type-Aware Failure Prediction for Chart and Diagram Reasoning in Vision-Language Models: Mid-Term Report

Yash More, Naman Singhal, Shrish Kadam, Sanjith Ganapathi

{yash.more, naman.s, shrish.kadam}@research.iiit.ac.in,

sanjith.ganapathi@students.iiit.ac.in

International Institute of Information Technology, Hyderabad

## Abstract

A vision-language model (VLM) that answers a chart question wrongly may have misread the figure or stated something the figure does not contain. We ask whether such failure types are distinguishable in the model’s internal state before it answers. We cached every layer of Qwen2.5-VL-7B and LLaVA-NeXT-7B on the 2,500 ChartQA test questions. A linear probe on the last prompt token predicts a wrong answer with AUROC 0.88 for Qwen and 0.80 for LLaVA, well above surface features (0.72 and 0.61), and most of the signal survives regressing those features out. Labelling all 1,495 errors showed that fabrication is almost absent from ChartQA (1 and 17 cases): the real errors are misreads and arithmetic slips, which probes separate (0.90 and 0.96), also within a single question type. To study fabrication we built synthetic bar charts that ask about missing categories. Neither model refuses, yet both encode that the category is missing (AUROC 0.96 to 1.00). Probes for misreads and for missing-bar fabrications are largely type-specific in Qwen but not in LLaVA, and on synthetic charts the position of a value relative to the gridlines predicts misreads as well as the probes do. The synthetic probes transfer poorly to real charts.<sup>1</sup>

## 1 Introduction

VLMs are increasingly used to read charts, plots and diagrams whose answers feed decisions. When such a model answers incorrectly, it can be wrong in two mechanically different ways. It may have **misread the figure**: picked the wrong bar, misjudged an axis tick, traced the wrong edge. The correct answer was in the image, and a more careful look might recover it. Or it may have **fabricated**: stated a value or label that appears nowhere in the figure. Looking again will not help, and the answer should be withheld.

Reliability tooling today collapses both into one “do not trust” verdict. Abstaining on everything discards recoverable answers; re-querying everything spends compute on answers that were never grounded. Our project asks whether the distinction is visible in the model’s internal state *before* generation, and whether knowing it is worth anything.

**Refined problem statement.** Let  $x = (I, q)$  be a chart image and a question, and  $\hat{y}$  the answer a frozen VLM generates greedily. Let  $h_\ell^p(x) \in \mathbb{R}^d$  be the model’s hidden state after decoder layer  $\ell$ , pooled at prompt position  $p$ , computed in the prefill pass that precedes the first answer token. Each wrong answer carries a failure type  $t \in \{\text{STRUCTURAL}, \text{FABRICATION}\}$  determined from  $\hat{y}$  and the figure. We test (i) whether  $\Pr(\hat{y} \text{ wrong} \mid h)$  and  $\Pr(t \mid h, \hat{y} \text{ wrong})$  can be estimated by linear probes on  $h_\ell^p$  better than by surface features of the input, and (ii) whether the probe for one failure type also responds to the other (cross-transfer). Later stages (E4, E5) test whether the type determines which repair works.

**Contributions.** (i) Full-depth activation caches for two VLMs on ChartQA and on synthetic charts, taken from the pass that produced each answer (Section 4); (ii) five-label annotations of every ChartQA error of both models; (iii) binary and type-aware probes, each reported against an activation-free baseline and checked over several data splits, with the surface features regressed out, and within a single question type (Section 5); (iv) a synthetic arm that induces fabrication, with controls that rule out the question text and the image alone (Section 6). Section 7 discusses the hypothesis and Section 8 the plan for the remaining weeks.

<sup>1</sup>Code and results: <https://github.com/the-neemon/type-aware-vlm-failures>

## 2 Background and Related Work

### 2.1 Pre-generation failure detection

Kogilathota et al. (2026) establish that hallucination risk in VLMs is readable from internal representations in a single forward pass, without decoding, reaching up to 0.93 AUROC across eight models. Two of their findings shaped our design: the most informative representation family varies by architecture, with Qwen2.5-VL-7B best served by visual-only features at roughly 0.79 AUROC, where most models rely on late query-token states. Zhang et al. (2026) localise the signal further, isolating hallucination-relevant attention heads through a variational information bottleneck and using probe gradients to intervene at inference time. Both works address binary object hallucination on natural photographs. Neither asks whether distinct kinds of failure are internally distinguishable.

### 2.2 Evidence that failure types differ mechanistically

Liu et al. (2026) apply activation patching across five VLMs and identify a visual grounding pathway supporting correct predictions alongside a separate hallucination pathway driving erroneous ones, the former at mid-to-late depths and the latter concentrated at early layers and network boundaries. The identified circuit transfers selectively to relational but not attribute hallucination. That selectivity is the closest existing evidence that hallucination is not one internal phenomenon, and is the empirical basis for our hypothesis. It also motivates probing across the full depth of the network, since the two pathways do not occupy the same layers.

### 2.3 Chart and diagram reasoning

ChartQA (Masry et al., 2022) remains the standard benchmark for chart question answering with visual and logical reasoning. Rani et al. (2025) contribute chart, question, reasoning-step and attribution annotations, identifying which regions justify a model’s answer. Iyengar et al. (2026) assemble diagram QA instances across six source datasets, annotating the minimal regions required to verify each answer, and find that correct answers do not guarantee a model grounded its reasoning in the supporting regions. Both resources annotate *where a model should look*. Neither labels *how a model failed*, which is the supervision signal we need.

### 2.4 Error taxonomies and visual self-correction

Ashury-Tahan et al. (2026) argue that benchmark accuracy reports whether a model fails but not why, and build an LLM-judge pipeline for taxonomy construction across 35 datasets and 83 models. Their setting is text-only LLMs; we borrow the methodology, not the taxonomy. Li et al. (2026) propose Visual Self-Refine, in which a chart-parsing model emits pixel-level localisations, visualises them, and feeds them back to itself to correct its own perception errors. Making a model re-examine a figure is therefore established; what is untested is whether the *appropriate* repair depends on which failure occurred.

### 2.5 Probe validity

Two results define our standard of evidence. Sahoo et al. (2026) report probes separating reasoning types at 100% accuracy with well-separated geometry, then show the separation was driven entirely by format confounds, collapsing to chance once source identity, option count and response length were residualised out. Yuan et al. (2026) report the healthy contrast, a hidden-state probe at 0.95 AUROC against a text-surface classifier at 0.59, but find that four interventions built on that signal failed: the signal was diagnostic, not causal. Two lessons follow: a probe result is uninterpretable without a surface baseline, and detection does not imply a usable intervention.

### 2.6 Positioning

Pre-generation detection is binary and targets natural images; mechanistic type-selectivity has been shown for object hallucination, not structured-image reasoning; chart resources annotate evidence regions, not failure types; and visual self-correction is applied uniformly. Our work sits in the intersection: failure-*type* prediction on structured images, and whether that type should determine the repair.

## 3 Hypothesis

**Type Separability Hypothesis.** The pre-generation internal state preceding a structural misreading is linearly distinguishable from the state preceding a fabrication, because the former is a precision-limited perceptual failure while the latter reflects generation ungrounded in the image. We test this rather than assume it: a misread bar height may be a higher-precision instance of the same per-

ceptual failure, and the design treats a collapse of the two classes as a reportable outcome.

## 4 Experimental Setup

### 4.1 Models and inference

We use two frozen open VLMs: Qwen2.5-VL-7B (Bai et al., 2025) and LLaVA-NeXT with a Mistral-7B language model (llava-v1.6-mistral-7b; Liu et al., 2024). ChartGemma, the stretch goal, is dropped. Every question is asked with the standard ChartQA suffix “Answer the question using a single word or phrase.”, decoded greedily (at most 32 new tokens), in fp16 with SDPA attention on two RTX 2080 Ti GPUs. Hardware, library versions and settings are frozen and recorded with every run, because a different GPU kernel can change the generated answer and decouple the cached activations from the answer that was labelled. Answers are scored by ChartQA relaxed accuracy (numeric answers within 5% of the gold; strings by exact match).

### 4.2 Activation caching

Activations are captured by forward hooks *during the same* generate() call that produced the answer, from its prefill pass, so every vector belongs to the labelled answer. Layer  $\ell$  is the residual stream after decoder block  $\ell$ . We cache every layer (Qwen: 28 layers,  $d = 3584$ ; LLaVA: 32 layers,  $d = 4096$ ) at four pooled positions: **vision\_mean** and **vision\_max** (mean and elementwise max over the image tokens), **query\_last** (the final prompt position, i.e. the state from which the first answer token is predicted) and **query\_mean** (mean over the text tokens of the prompt). Pooling is what makes full-depth caching affordable: the image contributes on average 584 tokens per question for Qwen and 2,212 for LLaVA. Caches are stored in fp16 (1.5 GB for Qwen and 2.0 GB for LLaVA on the ChartQA test set).

### 4.3 Probes

A probe is an  $L_2$ -regularised logistic regression on one pooled vector. For a chosen layer  $\ell$  and position  $p$ , the input  $h = h_\ell^p(x)$  is z-scored with the mean and standard deviation of the *training* items only, and the probe outputs a score  $s(x) = w^\top h + b$ , trained by minimising the logistic loss plus  $\frac{\lambda}{2}\|w\|^2$  (L-BFGS; the intercept is not penalised). There are  $d + 1$  parameters (3,585 for Qwen, 4,097 for LLaVA), which is more than the number of posi-

tive training examples in every task, so  $\lambda$  is selected from  $\{1, 10, 10^2, 10^3, 10^4\}$ . Probes are scored by AUROC: the probability that a randomly chosen positive item receives a higher score than a randomly chosen negative one. We train two kinds:

**Single-split sweep probes** (binary, and absent-versus-present on synthetic data). Items are split 60/20/20 by *figure* (hashed, seed 42), so no chart appears in two splits. Every (position, layer,  $\lambda$ ) cell is fitted on train and scored on validation; the best cell is chosen on validation only, and the test set is scored once.

**Cross-validated type probes** (the E3 matrices on synthetic data and on ChartQA’s labelled errors, where some classes are too small for a single held-out split). Items are assigned to five folds by figure. Within each outer fold, layer and  $\lambda$  are chosen by an inner cross-validation over the other four folds, and the probe then scores the held-out fold. Every item thus receives an out-of-fold score from a probe that never saw its chart. Position is fixed to query\_last in advance.

All intervals are 95% percentile intervals from a bootstrap that resamples whole figures (2,000 draws), because questions on the same chart are correlated. Every probe is reported next to a baseline that sees no activations. On ChartQA this is the *surface baseline*: the same probe on four features of the input (question source, question length, number of image tokens, numeric gold answer). Following Sahoo et al. (2026), we also report *residualised* probes, trained after the surface features are regressed out of the activations by least squares fitted on training rows only.

## 5 Naturalistic Arm: ChartQA

Qwen answers 87.3% of the 2,500 ChartQA test questions correctly (relaxed accuracy; 79.8% of the 1,250 human-written and 94.8% of the 1,250 machine-augmented questions, over 1,509 charts) and LLaVA 52.9% (44.6% and 61.1%), leaving 317 and 1,178 wrong answers.

### 5.1 Labelling the errors

The proposal planned two classes plus AMBIGUOUS. A pilot of 20 errors showed that this forced many items into AMBIGUOUS for reasons unrelated to the model, so the team adopted five labels: STRUCTURAL (the answer is explained by misreading the figure: wrong bar or series, misread digits, imprecise reading, miscounting), FABRICA-

| Label        | Qwen2.5-VL-7B |     | LLaVA-NeXT-7B |     |
|--------------|---------------|-----|---------------|-----|
|              | <i>n</i>      | %   | <i>n</i>      | %   |
| STRUCTURAL   | 71            | 22  | 583           | 49  |
| COMPUTATION  | 93            | 29  | 435           | 37  |
| NOT_AN_ERROR | 112           | 35  | 69            | 6   |
| AMBIGUOUS    | 40            | 13  | 74            | 6   |
| FABRICATION  | 1             | 0.3 | 17            | 1.4 |
| errors       | 317           |     | 1,178         |     |

Table 1: Labels for each model’s ChartQA test errors. NOT\_AN\_ERROR: Qwen 51 equivalent format, 38 valid reading, 23 wrong gold; LLaVA 39, 20, 10. AMBIGUOUS: Qwen 14 ambiguous question, 11 wrong gold (model also wrong), 12 other, 3 unreadable; LLaVA 24, 34, 15, 1.

TION (content the figure does not support), COMPUTATION (inputs read correctly, arithmetic or logic wrong), NOT\_AN\_ERROR (the answer is actually right: an equivalent format, a wrong gold answer, or a valid reading of an ambiguous question) and AMBIGUOUS. On ChartQA, STRUCTURAL and COMPUTATION are frequent enough to train type probes (Section 5.3). All 317 Qwen errors and all 1,178 LLaVA errors were labelled by the same LLM annotator (Claude), which saw only the chart image, question, gold answer and model answer, never the model’s identity, following the written rubric. Two checks bound its consistency: before labelling LLaVA it re-labelled 50 random Qwen errors blind and agreed on 47 (Cohen’s  $\kappa = 0.917$ ), and LLaVA batches labelled before the rubric was final were re-labelled blind, changing 55 of 1,178 labels. On the 48 questions where both models gave the identical wrong answer, the two models’ labels agree on 45. The planned two-person human agreement check ( $\kappa > 0.6$  on 200 items) has not yet been run.

Table 1 gives three findings. First, **fabrication is essentially absent from ChartQA for both models**: 1 of Qwen’s 317 errors and 17 of LLaVA’s 1,178, although LLaVA errs almost four times as often. ChartQA questions are written about content the chart contains, so a model rarely has occasion to invent content; its errors are misreads and arithmetic. Most of LLaVA’s apparently invented numbers turned out to be garbled printed values (486 for a printed 4,863), which are misreads. The naturalistic arm therefore cannot supply fabrication examples, which is the risk the synthetic arm was planned for. Second, **35% of Qwen’s scored errors are not errors** against 6% of LLaVA’s: most often the answer is correct in another format (“1:2”

|              | Probe                  | 5 splits       | Resid. | Surface |
|--------------|------------------------|----------------|--------|---------|
| Qwen, all    | 0.877 [.83,.92]        | .862 $\pm$ .02 | .803   | .715    |
| Qwen, clean  | <b>0.932</b> [.90,.96] | .910 $\pm$ .02 | .901   | .786    |
| LLaVA, all   | 0.798 [.76,.84]        | .801 $\pm$ .02 | .783   | .608    |
| LLaVA, clean | <b>0.819</b> [.78,.85] | .810 $\pm$ .01 | .798   | .636    |

Table 2: Binary probe (wrong vs. right) on ChartQA: test AUROC [95% CI] on the reported split; mean  $\pm$  s.d. over five figure splits; with the surface features regressed out (Resid.); and the surface-only baseline. Clean: without NOT\_AN\_ERROR items. All query\_last; Qwen layer 21 of 28, LLaVA 24 of 32.

for 0.5), the gold answer is wrong, or the question admits the model’s reading. A strong model’s residual errors are dominated by scoring artefacts; 34 of Qwen’s and 44 of LLaVA’s gold answers are wrong. Counting NOT\_AN\_ERROR answers as correct raises accuracy from 87.3% to 91.8% (Qwen) and from 52.9% to 55.6% (LLaVA). Third, **the models fail differently**: LLaVA’s largest class is STRUCTURAL (49%: garbled printed values, adjacent bars, gridline snapping such as 10,000 for a printed 11,767), and its error type depends on the question source: 409 of its 692 errors on human-written questions are COMPUTATION, while 391 of its 486 errors on augmented questions are STRUCTURAL.

Appendix A gives examples of each label. Only 22% of Qwen’s scored errors are figure misreads; nearly as many are arithmetic or reasoning slips, which the two-class taxonomy of the proposal did not provide for.

## 5.2 Predicting whether the answer is wrong (E1)

Table 2 reports the binary probe. A sanity probe first confirmed that each cache decodes what it plainly contains (human-written vs. augmented question: validation AUROC 1.000 for Qwen, 0.999 for LLaVA). For both models the probe clearly exceeds the surface baseline, with non-overlapping intervals. Removing Qwen’s 112 NOT\_AN\_ERROR items raises the probe from 0.877 to 0.932: part of what looked like probe error was label noise. For LLaVA, whose NOT\_AN\_ERROR items are 6% of its errors rather than 35%, the same step gives a smaller rise, from 0.798 to 0.819. The probe is not reading question style, which alone predicts errors at 0.68 for Qwen because 80% of its errors are on human-written questions: within human-written questions only it reaches 0.895 (Qwen, cleaned) and 0.850 (LLaVA). LLaVA’s

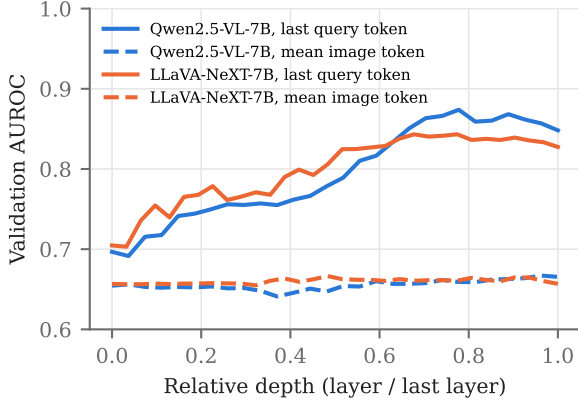


Figure 1: Binary probe, validation AUROC by relative depth (best  $\lambda$  per layer). Solid: last prompt token; dashed: mean over image tokens.

lower raw AUROC should not be read as a weaker signal; it is wrong on 47% of questions against Qwen’s 13%, so the two probes separate different error sets at different base rates, and its gap over its own baseline (+0.19) is similar to Qwen’s (+0.15).

**Robustness.** Over five random figure splits the probe moves by about  $\pm 0.02$ ; for Qwen the reported split lies above the mean (0.932 against 0.910 clean), so the mean is the fairer estimate. Regressing the surface features out of the activations lowers the probe by 0.02 to 0.07 but leaves it far above the surface baseline, so the signal is not a re-encoding of question style. Applied unchanged to 960 human-written ChartQA validation questions on charts never seen in training, Qwen’s probe scores 0.772 [0.73, 0.81], against 0.829 on the human-written test questions; adding these questions to training leaves test AUROC essentially unchanged (0.884 against 0.877). The probe’s probabilities are already calibrated (expected calibration error 0.03 for Qwen, 0.05 for LLaVA), which matters for a controller that thresholds them.

**Which layers.** Figure 1 shows validation AUROC by relative depth. In both models the signal is in the query-side state and builds with depth: query\_last rises from about 0.70 at the first layer to a plateau over the last third of the network (Qwen 0.85 to 0.87 over layers 18 to 27, best at 21; LLaVA 0.83 to 0.84 over layers 20 to 28, best at 24). The image-token positions stay flat at 0.62 to 0.68 at every depth, close to the surface baseline. This differs from HALP’s finding that Qwen is best served by visual-only features for object hallucination on photographs; for chart errors the informative state is the one that has already read the question.

| Probe (trained on)   | Qwen                   | LLaVA                  |
|----------------------|------------------------|------------------------|
| Misread vs. correct  | <b>0.892</b> [.85,.93] | <b>0.813</b> [.79,.83] |
| surface              | 0.651                  | 0.536                  |
| retrieval qs. only   | 0.913 (.692)           | 0.805 (.559)           |
| residualised         | 0.830                  | –                      |
| on comp. vs. corr.   | 0.820                  | 0.586                  |
| Comp. vs. correct    | <b>0.904</b> [.87,.93] | <b>0.949</b> [.94,.96] |
| surface              | 0.818                  | 0.821                  |
| arithmetic qs. only  | 0.845 (.553)           | 0.836 (.751)           |
| residualised         | 0.874                  | –                      |
| on misread vs. corr. | 0.732                  | 0.540                  |
| Misread vs. comp.    | <b>0.897</b> [.84,.94] | <b>0.961</b> [.95,.97] |
| surface              | 0.693                  | 0.845                  |
| residualised         | 0.918                  | –                      |
| classes (C / S / K)  | 2,183 / 71 / 93        | 1,322 / 583 / 435      |

Table 3: Type probes on ChartQA’s labelled errors (comp.: computation); out-of-fold AUROC [95% CI] on the contrast each probe was trained for; its surface-only baseline; the same probe trained and scored within one question type (surface baseline in brackets); with surface features regressed out (run for Qwen only); and applied to the other error type. Five folds by figure; query\_last; layers chosen per fold.

### 5.3 Error types on real charts (E3, naturalistic)

With 71 (Qwen) and 583 (LLaVA) misreads and 93 and 435 computation errors, the ChartQA labels support type probes for the two error types real charts produce, though not for fabrication. We use the cross-validated design of Section 4.3 and train three probes: misread vs. correct, computation vs. correct, and misread vs. computation (Table 3).

**Misreads are detectable on real charts**, well above the surface features, for both models (0.892 and 0.813). For Qwen this agrees with an earlier single-split probe (0.905 with 13 test positives), now measured on all 71. **Misreads and computation errors are separable** (0.897 and 0.961), above a surface baseline that is itself strong for LLaVA (0.845), because LLaVA’s computation errors fall mostly on human-written questions and its misreads on augmented ones. **In LLaVA the two probes are type-specific**: the misread probe barely responds to computation errors (0.586) and the computation probe barely to misreads (0.540). Qwen’s misread probe also flags computation errors (0.820), but the surface features alone reach 0.765 on that contrast, so much of this is question style rather than a shared error direction. Two further checks from the same matrix: LLaVA’s 69 NOT\_AN\_ERROR answers look like correct ones to its misread probe (0.490), as they should; and its

17 fabrications look more like misreads than like correct answers (0.752, [0.64, 0.85]), a descriptive result only.

**Question type.** Computation errors arise only on questions that ask for an operation (“difference”, “ratio”, “average”), and that wording is visible from the first layer: the computation probe at layer 0 already reaches 0.865 for LLaVA. To close this route we tagged every question as *arithmetic* or *retrieval* with a keyword rule fixed before any restricted probe was run (it tags 75% and 72% of computation errors as arithmetic and 93% of misreads as retrieval), and re-ran each probe within one type. Within arithmetic questions the computation probe still reaches 0.845 (Qwen) and 0.836 (LLaVA), well above surface features, and within retrieval questions the misread probe reaches 0.913 and 0.805. Part of the original signal was the question, but most of it is not. Regressing the surface features out of Qwen’s activations leaves all three probes between 0.83 and 0.92.

## 6 Synthetic Arm

### 6.1 Design

The synthetic arm makes fabrication happen on purpose, by asking about categories that are absent from the chart. Two design constraints come from earlier checks. (1) A probe comparing ChartQA misreads with synthetic fabrications would learn “real vs. synthetic chart”. (2) On synthetic data the question template alone predicts the intended failure type at AUROC 1.000, so a probe comparing one template with another can read the template. Every contrast is therefore made *within one question template*, and every outcome is scored from the model’s actual answer, never from what the question was designed to induce.

**Pilot 1** rendered 300 bar charts (matplotlib; 4 to 7 bars; six themes such as fruits or cities; values on 0–50 or 0–100 axes). Each chart is asked the same question twice, in identical words: once about a bar that is shown (“What is the value of Mango?”) and once about a plausible name from the same theme that is not (“... of Guava?”). **Pilot 2** (300 charts, 1,800 questions) added three changes after pilot 1. (i) Category names come in lookalike pairs (Dubai/Dublin, Eagle/Falcon, Lute/Flute); a chart shows at most one of a pair, and half the absent questions ask for the partner of a shown bar. (ii) Three question families, each asked about a shown and a missing name: *value*, *compare* (“Which is

|                   |               | Qwen    | LLaVA   |
|-------------------|---------------|---------|---------|
| Shown bar, right  | value (P1+P2) | 554/600 | 185/600 |
|                   | compare       | 300/300 | 209/300 |
|                   | neighbor      | 294/300 | 255/300 |
| Missing, value    | rejected      | 15      | 0       |
|                   | “0”           | 494     | 232     |
|                   | fabricated    | 91      | 368     |
| Missing, compare  | fabricated    | 37      | 101     |
| Missing, neighbor | fabricated    | 300     | 290     |

Table 4: Synthetic outcomes (pilots 1 and 2). Value rows pool both pilots (600 missing-name questions); compare and neighbor come from pilot 2 (300 each).

larger, Guava or Apple?”) and *neighbor* (“Which bar is immediately to the right of Guava?”). Naming any bar in reply to the last is a fabrication with no numeric escape. (iii) Harder charts: values never multiples of 5, axes up to 200, sparse ticks. Generation is deterministic, and both models were run on byte-identical charts.

Answers to missing names are scored as *rejected* (“None”), *fabricated* (any other number; naming the missing bar, its lookalike or any bar for neighbor questions), *zero* (“0”) or *picked-present* (naming the shown bar in a compare question). The team decided that “0” to a value question is a refusal expressed as a number, not a fabrication: no bar in these charts has zero height, so the model is not reporting something it read, and Qwen says “0” mostly when no bar resembles the asked name (141 of 157 unrelated names). It still counts as a wrong answer.

### 6.2 How the models answered

Table 4 summarises the answers. **Neither model refuses:** Qwen rejects 15 of 1,200 missing-name questions and LLaVA none. Qwen answers most missing values with “0”; when it does invent a number, 57 of its 76 pilot-2 inventions are the value of the lookalike bar, i.e. it binds the missing name to the nearest-looking bar that is present (asked for Dubai, it reports Dublin). LLaVA invents far more often (368 vs. 91), mostly round numbers near some bar. The models also differ on shown bars: Qwen reads 92% of values within tolerance, LLaVA 31%, because LLaVA snaps its readings to the nearest gridline (52 becomes 50). This yields four classes for the value template: C correct read, S misread (structural), F invented number (fabrication) and Z “0” (Qwen 554/46/91/494; LLaVA 185/415/368/232).

| Family               | Qwen        | LLaVA       |
|----------------------|-------------|-------------|
| value                | 0.999 (L4)  | 0.977 (L12) |
| compare              | 0.991 (L4)  | 0.959 (L19) |
| neighbor             | 1.000 (L5)  | 0.996 (L18) |
| lookalike names only | 0.980–1.000 | 0.922–0.991 |
| text-only baseline   | 0.49–0.52   | 0.49–0.52   |
| image-token probe    | 0.500       | 0.500       |

Table 5: Test AUROC of probes separating a missing from a shown name under identical wording (pilot 2; 62 charts per class in test). The image-token probe is a built-in control: image tokens precede the question, so they cannot see it.

### 6.3 Do the models know the bar is missing?

Table 5 shows that **both models represent that the asked name is missing, although they answer anyway**. Two controls make the result interpretable. A classifier on the question text alone is at chance, because every name is a shown bar on some charts and missing on others. Probes on the image-token positions score exactly 0.500: the image precedes the question in the prompt and attention is causal, so both questions of a chart receive identical image vectors. The signal holds for lookalike names, and it appears early in Qwen (validation 0.99 from layer 2) and later in LLaVA (about layer 9). The absence probe does not respond to misreads of shown bars (Qwen 0.46, LLaVA 0.51 on shown-bar value questions).

### 6.4 Type probes and cross-transfer (E3)

Table 6 is the central result. Each probe is trained on one pair of classes and then scored on every pair. If misreads and fabrications leave different internal signatures, each probe should be strong on its own task (bold) and weak on the other failure type.

**Qwen: largely type-specific.** The structural probe separates misreads from correct reads (0.887) but barely responds to fabrications (0.594); the fabrication probe is perfect on its own task and near chance on misreads (0.576, interval touching 0.5). The structural probe exceeds its metadata baseline by a modest 0.07 with overlapping intervals: much of Qwen’s misread risk is chart difficulty visible in the settings (0–200 axes with sparse ticks account for most misreads).

**LLaVA: the misread signal is not type-specific.** Its structural probe fires *more* on fabrications (0.879) and on “0” answers (0.987) than on its own task (0.776). It behaves like a general “this answer is not grounded” direction, strongest when

there is no bar at all.

**Gridline position explains much of the misread signal.** Relaxed accuracy tolerates a 5% *relative* error, so whether a coarse reading counts as a misread depends on how far the true value lies from a gridline relative to its size: rounding 34 to 30 is wrong, 134 to 130 is not. Two features declared before testing (the distance to the nearest gridline in gridline steps, and to the nearest multiple of 10), added to the metadata baseline, reach 0.853 for Qwen and 0.642 for LLaVA, below both probes. The same distances divided by the value, added after seeing that result, reach 0.920 and 0.860, above both probes (0.887 and 0.776). On synthetic charts, then, the misread probes cannot be told apart from a property of the chart and the scorer.

**In both models the “fabrication” probe is a missing-bar detector.** It scores “0” answers exactly as high as invented numbers (Z vs C 1.000) and even ranks them above (F vs Z below 0.3). What separates an invented number from “0” is decodable (0.984, 0.986), but only in the last layers (Qwen 22 to 27, LLaVA 23 to 30), where it largely reads out the answer about to be produced rather than giving an early warning.

**Transfer to real charts.** Trained on synthetic charts and applied unchanged to ChartQA, the misread probe barely separates real misreads from correct answers (Qwen 0.597 [0.53, 0.67], LLaVA 0.557 [0.53, 0.58]), and Qwen’s version responds more to real computation errors (0.791). The missing-bar probe scores LLaVA’s 17 real fabrications at 0.650 [0.52, 0.77], but real misreads just as high (0.669). The synthetic probes do not carry over to real charts, where probes trained on real labels work (Section 5.3). Layer profiles of the synthetic probes are in Appendix B.

## 7 Discussion

**Is the hypothesis supported?** Partly. Three findings support it. On real charts, misreads and arithmetic errors are linearly separable in both models (0.90 and 0.96), also within a single question type, and in LLaVA each type probe is nearly blind to the other type. On synthetic charts both models encode that the asked category is missing, with every control at chance. In Qwen, the synthetic misread probe barely responds to missing-bar fabrications, and the missing-bar probe barely to misreads. Two findings limit it. In LLaVA the synthetic misread probe fires more on fabrications than on misreads,

| Probe (trained on)         | Qwen2.5-VL-7B |              |        |              | LLaVA-NeXT-7B |              |        |              |
|----------------------------|---------------|--------------|--------|--------------|---------------|--------------|--------|--------------|
|                            | S vs C        | F vs C       | Z vs C | F vs Z       | S vs C        | F vs C       | Z vs C | F vs Z       |
| structural (S vs C)        | <b>0.887</b>  | 0.594        | 0.673  | 0.400        | <b>0.776</b>  | 0.879        | 0.987  | 0.215        |
| metadata baseline          | 0.814         | 0.557        | 0.515  | 0.543        | 0.512         | 0.522        | 0.486  | 0.537        |
| fabrication (F vs C)       | 0.576         | <b>1.000</b> | 1.000  | 0.286        | 0.680         | <b>0.988</b> | 1.000  | 0.243        |
| metadata baseline          | 0.588         | 0.840        | 0.564  | 0.787        | 0.642         | 0.769        | 0.643  | 0.651        |
| invented vs. zero (F vs Z) | 0.336         | 0.138        | 0.000  | <b>0.984</b> | 0.520         | 0.226        | 0.000  | <b>0.986</b> |

Table 6: Cross-transfer matrix on synthetic value questions (5-fold CV by chart, out-of-fold AUROC). Rows: the probe and what it was trained on; columns: the contrast it is scored on. Bold: each probe on its own task. Metadata baseline: the same procedure on chart settings only (pilot, axis range, tick density, bar count, lookalike present, phrasing). 95% CIs for the bold cells: Qwen [0.82, 0.94], [1.00, 1.00], [0.97, 0.99]; LLaVA [0.73, 0.82], [0.98, 0.99], [0.98, 0.99]; for the structural probe on F vs C, Qwen [0.55, 0.65], LLaVA [0.85, 0.90].

which is what the proposal called “one signal wearing two labels”. And on synthetic charts gridline position predicts misreads as well as either probe, so the synthetic misread signal may describe the chart rather than the model. Since the synthetic probes do not transfer to real charts, the cleanest evidence for type separability is the real-chart contrast between misreads and computation errors, a different pair of types from the one the proposal named.

**What the fabrication signal is.** With our data, a fabrication is defined by the missing category, so “predicting fabrication” and “detecting that the asked category is missing” cannot be separated. The finding we can state is that both models encode, early and linearly, that the question is about something not on the chart, and then answer it anyway. This is the signal a controller needs: it identifies exactly the questions where re-reading cannot help and abstention is the right repair.

**Validity.** Every number above sits next to a baseline with no access to activations: surface features on ChartQA (also regressed out of the activations), a text-only baseline and an image-token control on synthetic data, and a chart-metadata baseline for the type probes. The image-token control is exact by construction, which rules out leakage through the image. The ChartQA probes are stable across five figure splits and within a single question type.

## 8 Timeline for the Rest of the Project

The final submission is due on 31 October; the code for E4 is written and tested.

- **3–10 Oct.** E4, the cross-intervention matrix: every synthetic value-question error of both models (Qwen 631, LLaVA 1,015) re-asked with a question-conditioned crop, an

upsampled image, a “verify your evidence” prompt and a sampled re-ask, scored by the pre-registered difference-in-differences test (crop helps misreads more than missing-bar fabrications). In parallel: the two-person human agreement check on 200 errors from both models ( $\kappa > 0.6$ ), and the LLaVA residualised type probes.

- **11–20 Oct.** E5, the controller: abstain when the missing-bar probe fires, re-look when the binary or misread probe fires; risk–coverage curves and AURC against binary selective prediction and a label-aware oracle. In parallel: a synthetic pilot with out-of-range questions (e.g. a year the chart does not show), so that fabrication no longer coincides with a missing category.
- **21–31 Oct.** Final paper, code release and viva slides.

## 9 Conclusion

Before a VLM answers a chart question, a linear probe on its last prompt token predicts whether the answer will be wrong, beyond the question’s surface. On real charts the errors are misreads and arithmetic slips, which leave different internal signatures. Fabrication can be induced with questions about missing categories, and both models encode the absence yet answer anyway, a signal suited to deciding when to abstain.

## Limitations

The ChartQA labels come from one LLM annotator (blind self-agreement  $\kappa = 0.917$ ) and have not yet passed a two-person human check. Fabrication is nearly absent from ChartQA, so the misread–fabrication contrast rests on synthetic bar charts

and one question template, where fabrication coincides with a missing category, and the synthetic probes do not transfer to real charts. Qwen has only 46 synthetic and 71 real misreads. The question-type rule is a keyword heuristic, the residualised type probes cover Qwen only, and the best gridline features were chosen after seeing the data. The probes are diagnostic; whether they select better repairs or abstentions is untested.

## References

- Shir Ashury-Tahan et al. 2026. ErrorMap and ErrorAtlas: Charting the failure landscape of large language models. *arXiv preprint arXiv:2601.15812*.
- Shuai Bai et al. 2025. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*.
- Anirudh Iyengar Kaniyar Narayana Iyengar, Tampu Ravi Kumar, Gaurav Najpande, Manan Suri, Dinesh Manocha, Puneet Mathur, and Vivek Gupta. 2026. DRAGON: A benchmark for evidence-grounded visual reasoning over diagrams. *arXiv preprint arXiv:2604.25231*.
- Sai Akhil Kogilathota, Sripadha Vallabha E G, Luzhe Sun, and Jiawei Zhou. 2026. HALP: Detecting hallucinations in vision-language models without generating a single token. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6067–6085.
- Jinsong Li, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Jiaqi Wang, and Dahua Lin. 2026. Visual self-refine: A pixel-guided paradigm for accurate chart parsing. In *International Conference on Learning Representations (ICLR)*.
- Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. 2024. [LLaVA-NeXT: Improved reasoning, OCR, and world knowledge](#).
- Jiaxin Liu, Ding Zhong, Yue Wang, Zhidong Yang, Zhaolu Kang, Guangyuan Dong, Qishi Zhan, Pengcheng Fang, and Aofan Liu. 2026. Dual-pathway circuits of object hallucination in vision-language models. *arXiv preprint arXiv:2605.13156*.
- Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. ChartQA: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2022*.
- Anku Rani, Aparna Garimella, Apoorv Saxena, Balaji Vasan Srinivasan, and Paul Pu Liang. 2025. RADAR: A reasoning-guided attribution framework for explainable visual data analysis. *arXiv preprint arXiv:2508.16850*.
- Subramanyam Sahoo, Vinija Jain, Aman Chadha, and Divya Chaudhary. 2026. Linear probes detect task format, not reasoning mode in language model hidden states. *arXiv preprint arXiv:2606.02907*.
- Aojie Yuan, Zhiyuan Julian Su, Haiyue Zhang, Yi Nian, and Yue Zhao. 2026. Hidden error awareness in chain-of-thought reasoning: The signal is diagnostic, not causal. *arXiv preprint arXiv:2605.09502*.
- Feiran Zhang, Yixin Wu, Zhenghua Wang, Xiaohua Wang, Changze Lv, Xuanjing Huang, and Xiaoqing Zheng. 2026. VIB-probe: Detecting and mitigating hallucinations in vision-language models via variational information bottleneck. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23509–23521.

## A Examples of the error labels

Examples from Qwen’s labels, with the annotator’s rationale paraphrased:

- **STRUCTURAL** (71, 22%): a pie chart with seven labelled sectors counted as six; the adjacent 2017 bar reported instead of the asked 2018 bar; the top of a stacked segment reported instead of its height; a reversed comparison of two bars of similar length.
- **COMPUTATION** (93, 29%, the largest class of real errors): every input is printed and readable, but the operation is wrong, e.g. a ratio of levels (83/11) where the question asks for a ratio of changes, a sum over four printed values that matches no subset of years, or picking the larger of two printed changes wrongly.
- **NOT\_AN\_ERROR** (112, 35%): *format*: 0.95 for a gold of 95%, “increase” for “increasing”, or the same bar named by a different colour word; *gold wrong*: the model’s  $28/19 = 1.474$  is the correct ratio of the two bars while the gold 1.18 matches no ratio on the chart; *valid reading*: a question that names no year, answered for the latest year while the gold uses the first.
- **AMBIGUOUS** (40, 13%): both the gold and the model are wrong (11), the question has several readings and the model fits none (14), the needed value is unreadable, e.g. two near-identical shades of teal (3), or a misread cannot be told apart from an arithmetic slip (12, “other”).
- **FABRICATION** (1): pie slices of 35, 40, 13 and 11, where the answer 75 is a sum of two slices; the model’s 55 matches no slice and no sum.

## B Layer profiles of the synthetic type probes

Figure 2 shows own-task AUROC by relative depth. The missing-bar signal saturates early (Qwen by layer 3, LLaVA by about layer 8). The misread signal builds gradually and peaks in the last third (layers chosen by inner CV: Qwen 16 to 25, LLaVA 20 to 30).

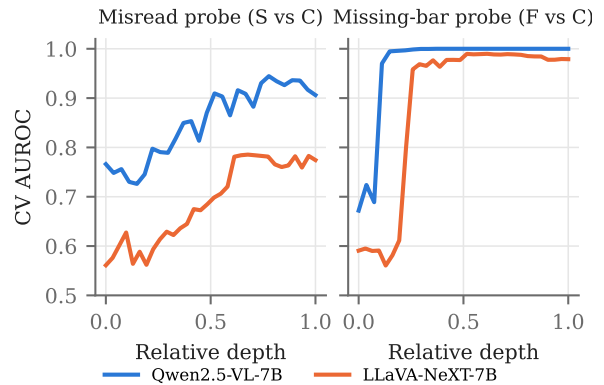


Figure 2: Type probes on synthetic value questions: out-of-fold AUROC on their own task by relative depth ( $\lambda$  chosen per fold by inner CV).